

Assignment 03

Handed out: 09/19/2026

Due by 1 PM EST on Saturday, 09/26/2026

The purpose of this assignment is to demonstrate the ability of Anthropic Claude Code and OpenAI to generate working code. As the template (objective) use the section entitled Building an Image Classifier with PyTorch in the attached notebook `10_neural_nets_with_pytorch.ipynb`. That notebook contains the code for chapter 10 of the book “Hands-On Machine Learning with Scikit-Learn and PyTorch” by Aurelien Geron (2025)

Problem 1. Use `Claude.ai/code` to incrementally build the whole process. Add code that will plot the training accuracy. Capture all generated scripts in your GitHub repository. Download the content of the repository to your local system and organize generated scripts in a Jupyter notebook. Please organize scripts as cells and place them in proper, executable order. Clearly label which cell in the notebook corresponds to which generated script. Ask `Claude.ai/code` to generate the summary of your dialog. Submit the notebook, its HTML image and the summary. (25%)

Problem 2. Without leaving the session, ask `Claude/code` to wrap the code in a Jupyter notebook. Download the notebook to your system and test it. Submit this notebook as well and its HTML image. (5%)

Problem 3. Start a new session. This time ask Claude/code in a longer single prompt to accomplish all it did in Problem 1 and 2. It is alright to ask for corrections (iterate). Submit newly generated notebook and its HTML image. (20%)

Problem 4. Accomplish the tasks performed in Problem 1, this time using OpenAI codex. (25%)

Problem 5. Accomplish the tasks performed in Problem 3, this time using OpenAI codex. (25%)

Formatting Requirements:

Solution of this assignment should preferably be submitted as Jupyter notebooks (ipynb files), as well as their HTML images. To generate the HTML image, use the following `jupyter` command:

```
$ jupyter nbconvert e89_LastName_FirstName_HW03_Prob#.ipynb --to html
```

Please provide meaningful description of actions in all cells of your notebook. Include those descriptions either as Python comments (`#`) or contents of the adjacent markdown cells.

Please name your notebooks `e89_LastName_FirstName_HW03_Prob#.ipynb`. Please replace Prob# with the actual Problem number it contains. We need your name in the file name and inside the file. Please add your name in a markup cell in the notebook itself. Claude/code and Codex can perhaps to both. Upload to Assignment 03 in Canvas.